

Logic and Critical Thinking

Informal Fallacies

Recap

- Arguments and Premises
- Sentence vs. proposition
- Truth, validity, and soundness
- Deductive and Inductive inferences
- Logical vs. physical impossibility

Significance of Deductive Inference

- **Deductive inference as**
 - **Non-ampliative**
 - It does not provide us with any new information
 - The truth of the conclusion is hidden within the premises
 - All reptiles lay eggs; no mammals lay eggs; no mammals are reptiles.
- **If the conclusion of a valid deductive argument turns out to be false, then it means one of the premises are false**
 - if you find a sad entrepreneur, then you call into question the truth of one of the premises in the valid argument:
 - all entrepreneurs are billionaires; all billionaires are happy; therefore, all entrepreneurs are happy.
 - or if you find a species that lays eggs as well as suckles its young, then either your premise that no reptiles lay eggs is wrong, or your definition of a ‘mammal’ or a ‘reptile’ is inadequate.

Inductive Inference

- **As ‘*ampliative*’**
 - It provides new information
- **They are non-monotonic**
 - Can always be made weaker or stronger by additional information
- **From a sample to a population**
 - every swan I have ever seen has been white therefore all swans are white
- **From a correlation to a causal generalization**
 - Everyone who is a chain smoker has a disposition to lung cancer. Smoking causes cancer.
 - Compare scientific generalizations with superstitions
 - RCTs

The circularity of Inductive Inference

- **What makes inductive arguments legitimate?**
 - **The Principle of the Uniformity of Nature**
 - Which tells us that the future will always be like the past
 - This underpins every single inductive argument
 - But what makes this principle legitimate?
 - Inductive inference
 - Hence, **circularity**
 - The future has always been like the past in the past
 - therefore the future will be like the past in the future

Truth, Validity, and Soundness: Various combinations

- valid arguments containing only true propositions
- valid arguments containing only false propositions
- invalid arguments containing only true propositions
- invalid arguments containing only true premises and have a false conclusion
- valid arguments having false premises and a true conclusion
- invalid arguments having false premises and a true conclusion
- invalid arguments, of course, containing all false propositions
 - See p. 30 of Copi and Cohen, *Introduction to Logic*

Exercise

- Many children have developed autism shortly after having been given the MMR jab, therefore the MMR jab causes autism.
- Many undergraduates like to get drunk at the weekend, so I should imagine that as it is Saturday, James may be in the pub.
- The IMD predicts thunderstorms over the next few days in Hyderabad. Therefore, there will be rain in the next few days.
- Since the 1950s, the atmospheric CO₂ level and the crime level have increased sharply. Atmospheric CO₂ causes crime.

Fallacies

- what is a fallacy?
 - A fallacy is a type of argument that **may seem to be correct, but that proves on examination not to be so.**
 - A form of argument in which the premises **appear to support** a conclusion but, in fact, do not; the term is often used to refer to familiar types of logical mistakes that may be used to trick or mislead.
 - A fallacy, also called a “courtroom trick” is an identifiable category of argument that does not support its conclusion.

Sophistry vs. Fallacy

- Sophistry is the art of deluding an audience with arguments that **one knows to be illogical or misleading**.
- On the other hand, when an illogical argument is **used innocently, without the speaker's being aware that it is**, in fact, illogical, then logicians call it a fallacy, a term that implies nothing about the speaker's motives.
- **Baseline check**
 - *Thank you for smoking* (2005)

Classification of Fallacies

- fallacies of relevance
- fallacies of defective induction
- fallacies of presumption
- fallacies of ambiguity

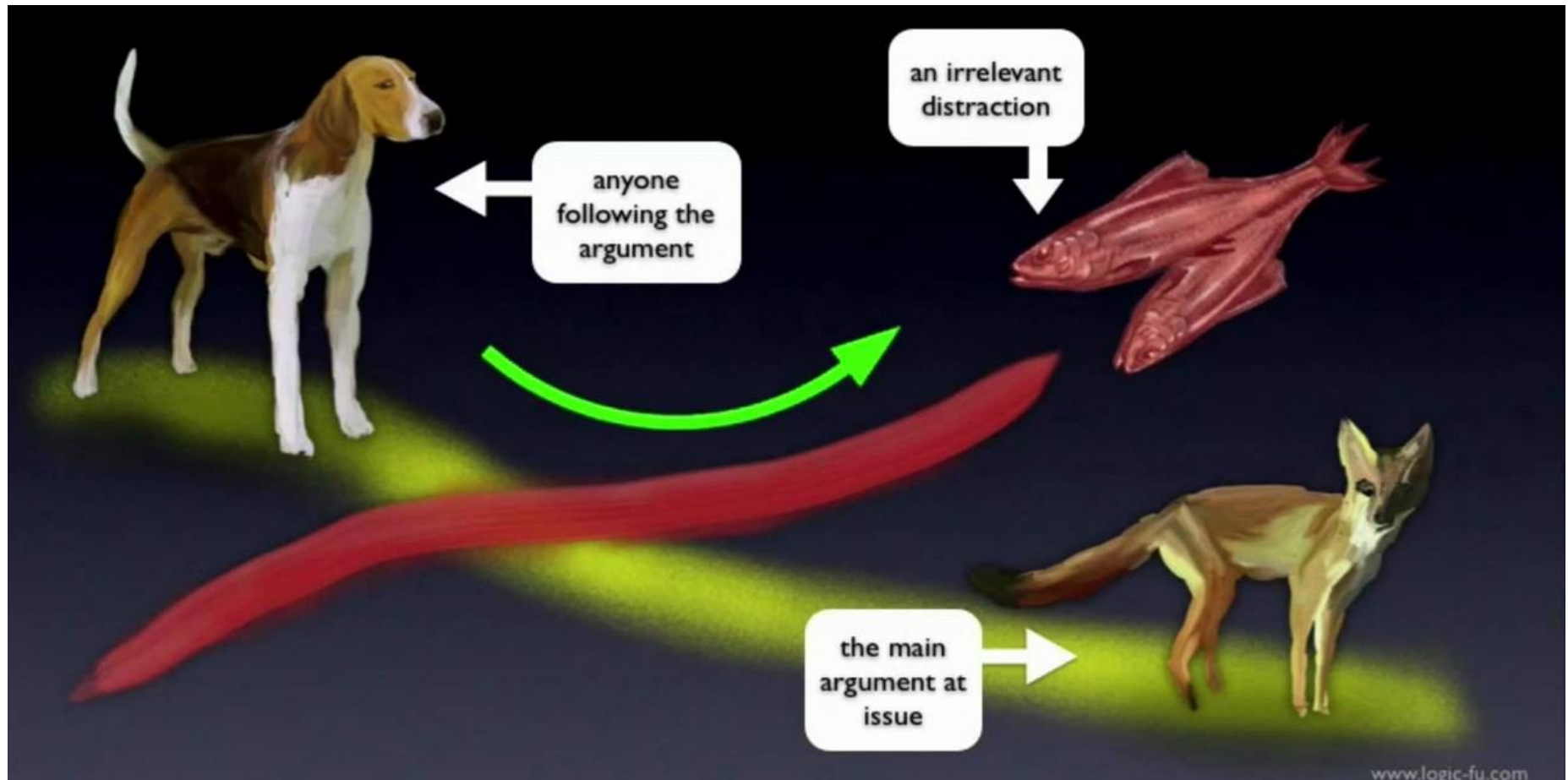
Fallacies of Relevance

- most numerous and frequent
- the **premises of the argument are simply not relevant to the conclusion.** However, because they are made to appear to be relevant, they may deceive
- they arise when there is **no real connection between the premises and the conclusion of an argument.** Because that connection is missing, **the premises offered cannot possibly establish the truth of the conclusion drawn.**

Fallacies of Relevance

- **The appeal to the populace**
- **The appeal to emotion**
- **The red herring**
- **The straw man**
- **The attack on the person (Ad hominem)**
 - Abusive
 - Circumstantial
 - Tu quoque
- **The appeal to force**
- **Missing the point (irrelevant conclusion)**
 - *Caveat...

A red herring



Red Herring

- when the arguer **diverts the attention of the reader or listener by changing the subject** to a different but sometimes subtly related one. He or she then finishes by either drawing a conclusion about this different issue or by merely presuming that some conclusion has been established.
 - Aka whataboutism
- Environmentalists are continually harping about the dangers of nuclear power. Unfortunately, electricity is dangerous no matter where it comes from. Every year hundreds of people are electrocuted by accident. Since most of these accidents are caused by carelessness, they could be avoided if people would just exercise greater caution.

Straw Man

- It is very much easier to win a fight against a person made of straw than against one made of flesh and blood. If one argues against some view by presenting an opponent's position as one that is easily torn apart, the argument is fallacious, of course. Such an argument commits the fallacy of the straw man.
 - Watch out for phrases such as “so what you’re saying is...” or “the real reason...”
- Principle of charity

Kinds of Ad Hominem

- **Abusive**

- “Television entertainer Bill Maher argues that religion is just a lot of foolish nonsense. But Maher is an arrogant, shameless, self-righteous pig. Obviously his arguments are not worth listening to.”
- **Character assassination**
- **Guilt by association**
- **Poisoning the well**
 - Attacking motivations

- **Circumstantial**

- **Tu quoque**
 - You did it too!

Missing the point

- This fallacy occurs when the premises of an argument support one particular conclusion, but then a different conclusion, often vaguely related to the correct conclusion, is drawn.
 - Crimes of theft and robbery have been increasing at an alarming rate lately. The conclusion is obvious: We must reinstate the death penalty immediately.
 - Abuse of the welfare system is rampant nowadays. Our only alternative is to abolish the system altogether.

Red herring vs. Straw man vs. Missing the point

- In the **straw man**, the arguer begins by distorting an opponent's argument and concludes by **knocking down the distorted argument**.
- In the **red herring**, the arguer ignores the opponent's argument (if there is one) and **subtly changes the subject**.
- Thus, to distinguish the two fallacies, one should attempt to determine **whether the arguer has knocked down a distorted argument or simply changed the subject**.
- Both the red herring and straw man fallacies are susceptible of being confused with missing the point, because all three involve a similar kind of irrelevancy. To avoid this confusion, one should note that **both red herring and straw man proceed by generating a new set of premises, whereas missing the point does not**.

Caveat!

- There is a sense in which every fallacy of irrelevance is an instance of missing the point (*ignoratio elenchi*), because in all these fallacies there is a gap between the premises and the conclusion. Premises that are not relevant—red herrings, straw men, personal attacks—all miss the point; that is true. **But we reserve this name for those fallacies of irrelevance that do not fit into other categories.** The *ignoratio elenchi* is, we may say, a catchall class of fallacies: fallacies in which the premises simply fail to connect to the intended conclusion with the coherence that rational argument requires.

Examples

- A majority of the population thinks we coddle criminals. It's time we got tough on crime.
- UFOs must be real because a significant portion of the population takes UFO sightings seriously.
- “Which phone should I go for?” “You must buy a Redmi phone, because it is the largest selling phone in India.”
- “Your gun control proposal is an affront to gun owner rights. We are talking about liberty being seized by an overinvasive government here.”
 - “Oh, that's funny coming from the person who proposed such draconian abortion regulations. If you want to talk about rights and liberty being stripped by an overinvasive government, there is example A.”

- The Republicans say we should shrink the deficit and cut down on government spending. But these are the same people who cranked up the deficit by getting us into two wars simultaneously. These are the same people who cranked up the deficit by passing tax breaks for the rich. These are the people who gave us the Bush years of spending more massive than any preceding administration of either party.
- Mr. Goldberg has argued against prayer in the public schools. Obviously Mr. Goldberg advocates atheism. But atheism is what they used to have in Russia. Atheism leads to the suppression of all religions and the replacement of God by an omnipotent state. Is that what we want for this country? I hardly think so. Clearly Mr. Goldberg's argument is nonsense.

- The United States says that North Korea is stockpiling nuclear weapons and is, therefore, a danger to world peace. But surely both Israel and the United States have their own stockpiles of nuclear weapons!
- “Kejriwal talks about the economy because it will benefit him politically.”
- Jamie Bartlett: “i think it’s an important job for journalists to try to ask about the negative possibilities of this stuff (wrt automation and disruption)”.

Sam Altman: “I think if you continue this thrust of, shouldn’t we stop progress, no one’s going to take you seriously, because people want this stuff”.

- You say that the changes to the tax code would promote fairness, but the real reason you are in favor of it is that you want to punish the rich.

- Look at the line for that new movie! It is out of the theater door, down the street, and around the corner. That film must be great.
- I was reading the expert reviews for this product on its website. All of them were outstanding. I think that this could be the product we've been waiting for.
- There's no point going to him for advice on your marriage. He's a divorcee.
- Philosophers will obviously argue against the possibility of AI, since it puts their whole profession at the risk of being replaced by more intelligent and conscious machines.

- You say caste-based reservation is way to correct the historical wrongdoings of our caste society. Then what about the economically poor who are from upper castes? Should we just ignore them?
- Those who resist [affirmative action programs] deny that they are racists, but the truth is that their real motivation is racism, a belief in the inherent inferiority of African-Americans and people of mixed racial backgrounds.
- Crimes of theft and robbery have been increasing at an alarming rate lately. The conclusion is obvious: We must reinstate the death penalty immediately.
- Abuse of the welfare system is rampant nowadays. Our only alternative is to abolish the system altogether.
- You say that we need to help the homeless, but what about the working poor who have a place to live? Do you think we should just ignore them?
- You know that famous celebrity who is always going on and on about the need to care about the environment and leave a small carbon footprint? It turns out that she has a mansion, and you know that thing uses a ton of electricity in the summer when she runs the air conditioner. So, if she can use a lot of power, so can I.

This good-old chestnut

- **A:** *“You really need to clean those dishes in the sink. You make yourself a snack and just clutter the kitchen and leave it for me. That is not respectful or fair to me.”* **B:** *“Well, if you want to talk about messes and respect, what about the fact that you never put out the garbage? You just let it get to a point where I have to always put it out myself.”*

How to discover a fallacy

- Identify the conclusion of the argument
- Identify the premises stated in support of the argument
- Do the premises lend independent support to the conclusion?
- No?? Then, it's a fallacy!
 - To identify the fallacy, ask *how* the premises is being used to support the conclusion.
 - Is it restating the conclusion in other words?
 - Is it setting up a false dichotomy?
 - Is it attacking the person/character/morality?
 - Is it playing with words?
 - Is it setting up a strawman version of the conclusion?
 - Is it distracting from the issue at hand?

- “Like any rational entity, the algorithm learns how to modify the state of its environment—in this case, the users mind—in order to maximize its own reward. The consequences include the resurgence of fascism, the dissolution of the social contract that underpins democracies around the world, and potentially the end of the European Union and NATO. Not bad for a few lines of code, even if it had a helping hand from some humans. Now imagine what a really intelligent algorithm would be able to do.”
 - Stuart Russell, *Human Compatible: Artificial Intelligence and the Problem of Control*

- “Here are a few examples of jobs that humans used to do but that technology automated long ago, at least in developed countries: clothes washer; rickshaw driver; elevator operator; punkawallah (a servant in India whose sole job was to work a manual fan for cooling the room, before the days of electric fans); computer (a human, usually female, who performed tedious calculations by hand, particularly during World War II). Most people will agree that in those instances replacing humans with machines in such jobs made life better all around. One could argue that today’s AI is simply extending that same arc of progress: improving life for humans by increasingly automating the necessary jobs that no one wants to do.”
 - Melanie Mitchell, Artificial Intelligence: A guide for thinking humans

Note about Refutation

- **Just because someone makes a bad argument for a conclusion does not give us rational justification for thinking that the conclusion is not the case.**
 - An argument provides us good reason to believe in the likely truth of the conclusion. **If an argument fails, then it means that those specific premises do not give us reason to believe that conclusion.**
- But it doesn't mean that there can be no other set of premises that does. **There could be a good argument for that conclusion, and this one just isn't it.** Every true proposition can be made the conclusion of a terrible argument.
 - All men are mortal. Thursday follows Wednesday; therefore New Delhi is the capital of India

Explore

- Peter Burke, *The Polymath: A cultural history from Leonardo da Vinci to Susan Sontag*
- David Epstein, *Range: Why generalists triumph in a specialized world*
- Waqas Ahmed, *The Polymath: Unlocking the power of human versatility*
- Tom Nichols, *The Death of Expertise: The Campaign against Established Knowledge and Why It Matters*

Practice

- E.g., *Wag the Dog* (1997)
- *Thank You for Smoking* (2005)
 - <https://www.youtube.com/watch?v=xuaHRN7UhRo>
 - Catch the fallacies?

Resources

- Copi and Cohen, *Introduction to Logic*
- Patrick Hurley, *A Concise Introduction to Logic*
- Edward T. Damer, *Attacking Faulty Reasoning: A Practical Guide to Fallacy-Free Arguments*
- Fallacyfiles.org
- <https://yourlogicalfallacyis.com/>
- <https://www.ancientaliensdebunked.com/>
- Podcasts
 - The Fallacy-a-Day Podcast
 - Fallacious Trump
- Watch
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